



ALMA MATER STUDIORUM
UNIVERSITÀ DI BOLOGNA
CAMPUS DI FORLÌ

91258 / B0385 Natural Language Processing

Lesson 5. Naïve Bayes' Classifier

Alberto Barrón-Cedeño
a.barron@unibo.it

12/10/2026

Previously

- One rule-based system for sentiment analysis
- Brief introduction to ML
- Foreword to Naïve Bayes

Naïve Bayes

Naïve Bayes

1. Introduced in the IR community by Maron (1961)
2. First machine learning approach
3. It is a **supervised** model
4. It applies Bayes' theorem with strong (naïve) independence assumptions between the features:
 - they are independent
 - they contribute "the same"

Naïve Bayes

A conditional probability model

Given an instance represented by a vector

$$\mathbf{x} = (x_1, \dots, x_n) \quad (1)$$

representing n **independent** features $x_1, x_2, x_3, \dots, x_{n-2}, x_{n-1}, x_n$
 n could be $|V|$ (the size of the vocabulary)¹

The model assigns to instance $\mathbf{x}$ the probability

$$p(C_k | \mathbf{x}) = p(C_k | x_1, \dots, x_n) \quad (2)$$

for each of the k possible outcomes C_k

where $C_k = \{c_1, \dots, c_k\}$

From https://en.wikipedia.org/wiki/Naive_Bayes_classifier

¹ $|\mathbf{x}|$ is the cardinality of $\mathbf{x}$

Naïve Bayes

Using Bayes' Theorem

The conditional probability $p(C_k | x_1, \dots, x_n)$ can be decomposed as

$$p(C_k | \mathbf{x}) = \frac{p(C_k) p(\mathbf{x} | C_k)}{p(\mathbf{x})} \quad (3)$$

Which can be read as

$$\text{posterior} = \frac{\text{prior} \times \text{likelihood}}{\text{evidence}}$$

But $p(\mathbf{x})$ does not depend on the class (constant $\forall$ classes):

$$p(C_k | \mathbf{x}) \sim p(C_k) p(\mathbf{x} | C_k) \quad (4)$$

From https://en.wikipedia.org/wiki/Naive_Bayes_classifier

Naïve Bayes

Going deeper (assuming a binary classifier)

$$p(C | \mathbf{x}) = \frac{p(C) p(\mathbf{x} | C)}{p(\mathbf{x})} \quad (5)$$

$$\text{posterior probability} = \frac{\text{class prior probability} \times \text{likelihood}}{\text{predictor prior probability}}$$

$p(C | \mathbf{x})$ Posterior probability of the class given the input²

```
if p > 0.5:
    class = positive
else:
    class = negative
```

Naïve Bayes

Going deeper (assuming a binary classifier)

$$p(C | \mathbf{x}) = \frac{p(C) p(\mathbf{x} | C)}{p(\mathbf{x})} \quad (6)$$

$$\text{posterior probability} = \frac{\text{class prior probability} \times \text{likelihood}}{\text{predictor prior probability}}$$

$p(C)$ Class **prior** probability
How many **positive** instances I have seen (during training)³

³Think of a bag marbles of different colours.

Naïve Bayes

Going deeper (assuming a binary classifier)

$$p(C | \mathbf{x}) = \frac{p(C) p(\mathbf{x} | C)}{p(\mathbf{x})} \quad (7)$$

posterior probability = $\frac{\text{class prior probability} \times \text{likelihood}}{\text{predictor prior probability}}$

$p(\mathbf{x} | C)$ Likelihood
The probability of the document given the class

Rough Idea

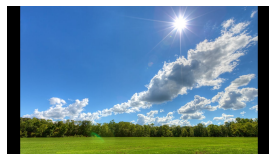
- The value of a particular feature is **independent** of the value of any other feature, given the class variable
- All features contribute the same to the classification
- Naïve Bayes' tries to find *keywords* in a set of documents that are predictive of the target (output) variable
- The internal coefficients will try to map tokens to scores
- Same as VADER, but without manually-created rules
the *machine* will estimate them

From (Lane et al., 2019, p. 65–68)

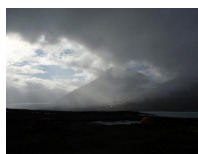
Naïve Bayes

A toy example: Should I ride my bike today?

One single factor



sunny



overcast



rainy



(here come some dense slides)

Naïve Bayes

A toy example: Should I ride my bike today?

Dataset	
Input	🚲
☀️	yes
☁️	yes
☀️	no
☔️	yes
☀️	yes
☁️	yes
☀️	yes
☔️	yes
☀️	no
☔️	no
☁️	yes
☔️	no
☔️	no

Computing **all** the probabilities by “counting”

Frequency table

🚲		
Input	yes	no
☀️	3	2
☁️	4	0
☔️	2	3

Likelihood table

🚲		
Input	yes	no
☀️	3/9	2/5
☁️	4/9	0/5
☔️	2/9	3/5

Naïve Bayes

A toy example: Should I ride my bike today?

Likelihood table

Input		
	yes	no
	3/9 ¹	2/5
	4/9	0/5
	2/9	3/5
	9/14 ²	5/14

$$^1 p(x | c) = p(\text{sun} | \text{yes}) = 3/9 = 0.33$$

$$^2 p(c) = p(\text{yes}) = 9/14 = 0.64$$

$$p(x) = p(\text{sun}) = 5/14 = 0.36$$

What is the Naïve Bayes' probability of yes if sun?

$$p(c | x) = p(c)p(x | c)/p(x)$$

$$p(\text{yes} | \text{sun}) = p(\text{yes})p(\text{sun} | \text{yes})/p(\text{sun})$$

$$p(\text{yes} | \text{sun}) = 0.64 * 0.33 / 0.36$$

$$p(\text{yes} | \text{sun}) = 0.59$$

Adapted from http://www.saedsayad.com/naive_bayesian.htm

Naïve Bayes

A toy example: Should I ride my bike today?

If...  let's ride !

Naïve Bayes

A toy example: Should I ride my bike today?

Considering more data

Sky	Temp	Humidity	Windy	
	hot	high	false	no
	hot	high	true	no
	hot	high	false	yes
	mild	high	false	yes
	cool	normal	false	yes
	cool	normal	true	no
	cool	normal	true	yes
	mild	high	false	no
	cool	normal	false	yes
	mild	normal	false	yes
	mild	normal	true	yes
	mild	high	true	yes
	hot	normal	false	yes

Adapted from http://www.saedsayad.com/naive_bayesian.htm

Naïve Bayes

A toy example: Should I ride my bike today?

Frequency tables

Sky	yes	no
	3	2
	4	0
	2	3
Humidity	yes	no
high	3	4
normal	6	1
Temp	yes	no
hot	2	2
mild	4	2
cool	3	1
Windy	yes	no
false	6	2

Likelihood tables

Sky	yes	no
	3/9	2/5
	4/9	0/5
	2/9	3/5
Humidity	yes	no
high	3/9	4/5
normal	6/9	1/5
Temp	yes	no
hot	2/9	2/5
mild	4/9	2/5
cool	3/9	1/5
Windy	yes	no
false	6/9	2/5

Adapted from http://www.saedsayad.com/naive_bayesian.htm

Naïve Bayes

Likelihood tables

Sky	yes	no
	3/9	2/5
	4/9	0/5
	2/9	3/5

Humidity	yes	no
high	3/9	4/5
normal	6/9	1/5

Temp	yes	no
hot	2/9	2/5
mild	4/9	2/5
cool	3/9	1/5

Windy	yes	no
false	6/9	2/5
true	3/9	3/5

Sky temp humidity windy ride
 cool high true ?

$$\begin{aligned}
 p(\text{yes} | x) &= \frac{p(\text{yes})p(\text{rain cloud} | \text{yes})p(\text{cool} | \text{yes})p(\text{high} | \text{yes})p(\text{true} | \text{yes})}{p(\text{rain cloud})p(\text{cool})p(\text{high})p(\text{true})} \\
 &= \frac{9/14 \times 2/9 \times 3/9 \times 3/9 \times 3/9}{5/14 \times 4/14 \times 7/14 \times 6/14}
 \end{aligned}$$

Adapted from http://www.saedsayad.com/naive_bayesian.htm

Naïve Bayes

Back to the definition...

$$p(c | x) \propto p(c)p(x | c) \quad (11)$$

Remember that x is a vector

$$p(c | x_1 \dots x_n) \propto p(c)p(x_1 | c) \times p(x_2 | c) \times \dots \times p(x_n | c) \quad (12)$$

Eq. (12) can be rewritten as

$$p(c | x_1 \dots x_n) \propto p(c) \prod_{i=1}^n p(x_i | c) \quad (13)$$

Naïve Bayes

Back to the definition...

$$p(C | x) = \frac{p(C) p(x | C)}{p(x)} \quad (8)$$

The probability $p(x)$ is constant for any given input

$$p(C | x) = \frac{p(C) p(x | C)}{p(x)} \quad (9)$$

$$p(c | x) \propto p(c)p(x | c) \quad (10)$$

Naïve Bayes

The classification process

Back to the toy example

$$\begin{aligned}
 p(\text{yes} | x) &\propto p(\text{yes})p(\text{rain cloud} | \text{yes})p(\text{cool} | \text{yes})p(\text{high} | \text{yes})p(\text{true} | \text{yes}) \\
 &\propto 9/14 \times 2/9 \times 3/9 \times 3/9 \times 3/9 \\
 &\propto 0.00529, \text{ which is not a probability}
 \end{aligned}$$

Classification: the maximum for all the classes

$$c \propto \arg \max_c p(c) \prod_{i=1}^n p(x_i | c) \quad (14)$$

```

compute p(yes | x)
compute p(no | x)
if p(yes | x) > p(no | x):
    yes
else:
    no

```

Training a Machine Learning Model

The dataset

We need a bunch of items (documents) with their associated **class**

kind	examples
binary	{positive, negative}
	{0, 1}
	{-1, 1}
multiclass	{positive, neutral, negative}
	{0,1,2}

In our case, we need the sentiment:

d_1	pos	d_5	neg	d_9	neu
d_2	neu	d_6	neg	d_{10}	pos
d_3	pos	d_7	neg	d_{11}	neu
d_4	pos	d_8	pos	d_{12}	neg

The dataset

Option 1 Use a corpus created by somebody else

Option 2 Build your own corpus⁴

- (a) You have/hire experts to do it
- (b) You engage non-experts through gamification
- (c) You hire non-experts through explicit crowdsourcing
- (d) There are many other ways to get annotated data

⁴Try to avoid it for this lesson.

Let us go and build a classifier with a corpus built by Hutto and Gilbert (2014)⁵

If you are following the **NLP in Action** book, they instruct you to download and install their software companion:

<https://github.com/totalgood/nlpia>

⁵<http://comp.social.gatech.edu/papers/icwsm14.vader.hutto.pdf>

What I did on OsX and GNU Linux

I use pipenv⁶

```
$ pipenv install --skip-lock nlpia
```

On Github they explain how to install it with conda or pip if you plan to contribute to the project

BUT, I modified the notebook to avoid the dependency on nlpia
(all it does is downloading the dataset!)

 Let us see it working
⁶<https://pipenv.readthedocs.io/en/latest/>

References

Hutto, C. and E. Gilbert

2014. VADER: A parsimonious rule-based model for sentiment analysis of social media text. In *Eighth International Conference on Weblogs and Social Media (ICWSM-14)*, Ann Arbor, MI.

Lane, H., C. Howard, and H. Hapkem

2019. *Natural Language Processing in Action*. Shelter Island, NY: Manning Publication Co.

Maron, M.

1961. Automatic indexing: An experimental inquiry. *Journal of the ACM*, 8:404–417.